Google Cloud H2S

AI Builder Cup

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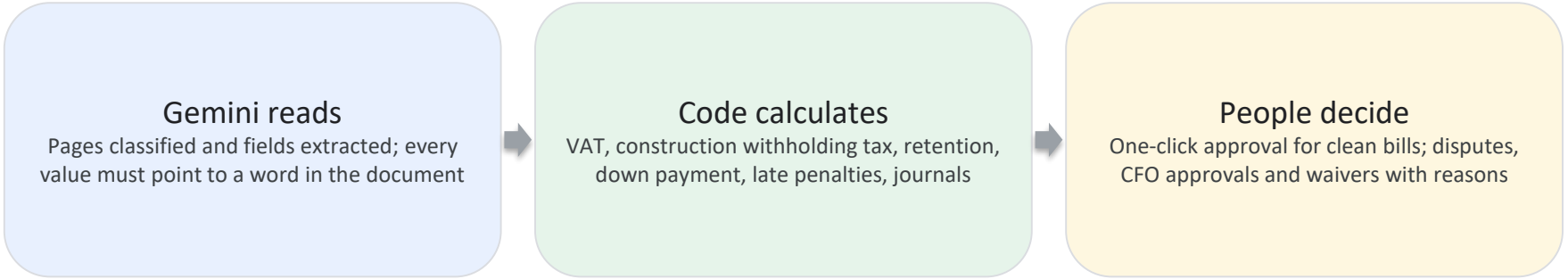
Team Details

a. Team name: Team OmniAudit
b. Team leader name: Rama Raditya
c. Problem Statement: Future of Work & Enterprise Productivity: finance teams at multi-entity property developers audit contractor bills and match buyer payments by hand

Brief about the idea

OmniAudit AI is an AI audit co-pilot for property developers that run every project as a separate company (SPV).

It checks each contractor bill before money goes out and matches each buyer payment to its unit when money comes in. Gemini reads the documents, deterministic code calculates every rupiah, and people make the decisions. Clean items are approved in one click; suspicious ones are locked and opened in a split-view audit screen, and every action lands in a tamper-evident audit trail.



Opportunities

Large developers process thousands of contractor bills and tens of thousands of buyer transfers a month. Overbilling, missed retention or penalties, and unallocated receipts leak real money and take days to check by hand.

How different is it from other ideas?

Generic invoice OCR stops at extraction. OmniAudit grounds every AI value in the source document and runs Indonesian construction rules (PP 9/2022 withholding by contractor grade, Coretax VAT base, retention, down payment, penalties) in deterministic code, across many SPVs at once.

How will it solve the problem?

Eight automatic checks lock risky bills before payment; a match score assigns transfers to units; the CFO gets one queue for large or suspicious cases; every decision is hash-chained.

USP

AI never calculates money. Results are explainable to an external auditor: each number links to its location in the PDF.

List of features offered by the solution

Payables (contractor bills)

- Upload a multi-page PDF: invoice, inspection report (BAP), tax invoice, photos
- Local OCR (digital and scanned) + Gemini extraction with grounding
- Contract agent: links the bill to its contract, vendor grade and terms
- 8 anomaly checks: overbilling, duplicate across SPVs, inflated amount, arithmetic, missing retention, wrong tax rate, tax-invoice mismatch, invalid progress
- Two-column calculation: claimed vs inspection-verified, to the rupiah
- Split-view audit screen with highlight boxes and click-to-jump evidence
- Overrides with mandatory reasons; balanced journal posted to ERP

Receivables (buyer payments)

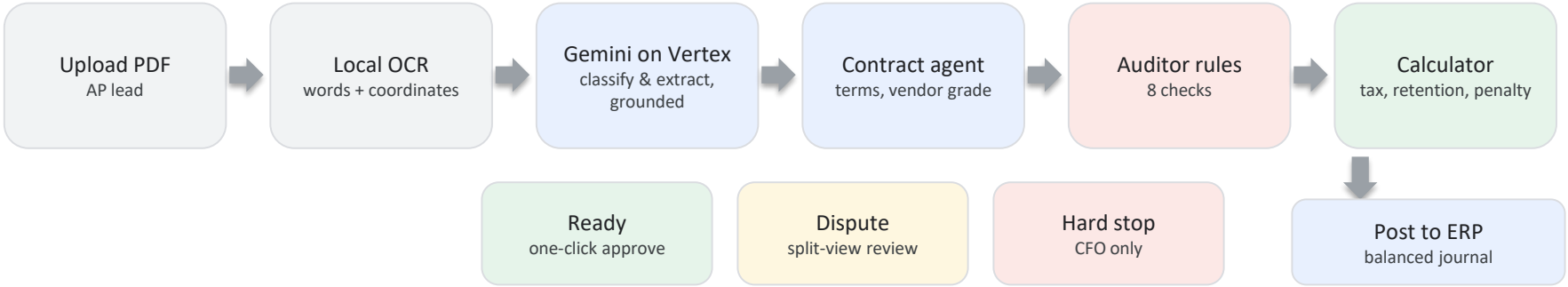
- Bank statement CSV upload; exact virtual-account matches auto-allocated
- Match score: 45% unit code + 35% payer name + 20% amount
- Late fee 0.1%/day per installment; waivers with reasons, executive waivers to CFO
- FIFO allocation, buyer credit, bank-fee tolerance
- Receipt PDF + WhatsApp/email notification (simulated)

Control & governance

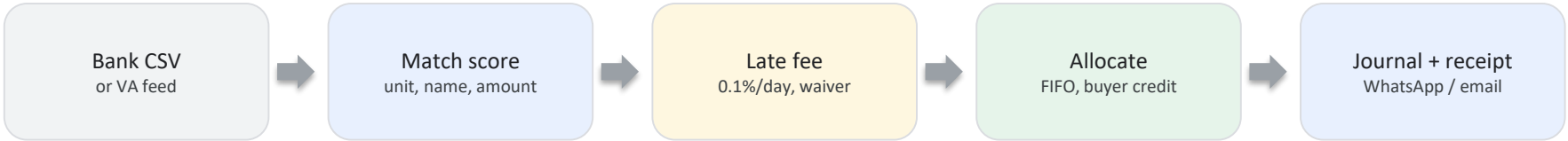
- Role-based access and data isolation per SPV
- CFO room: approvals above IDR 500 million, hard stops
- Hash-chained audit trail with integrity check
- AI accuracy tracked per field from human corrections
- Dashboard: unallocated payments, pending bills, disputes, leakage prevented

Process flow diagram or Use-case diagram

Contractor bill (payables)

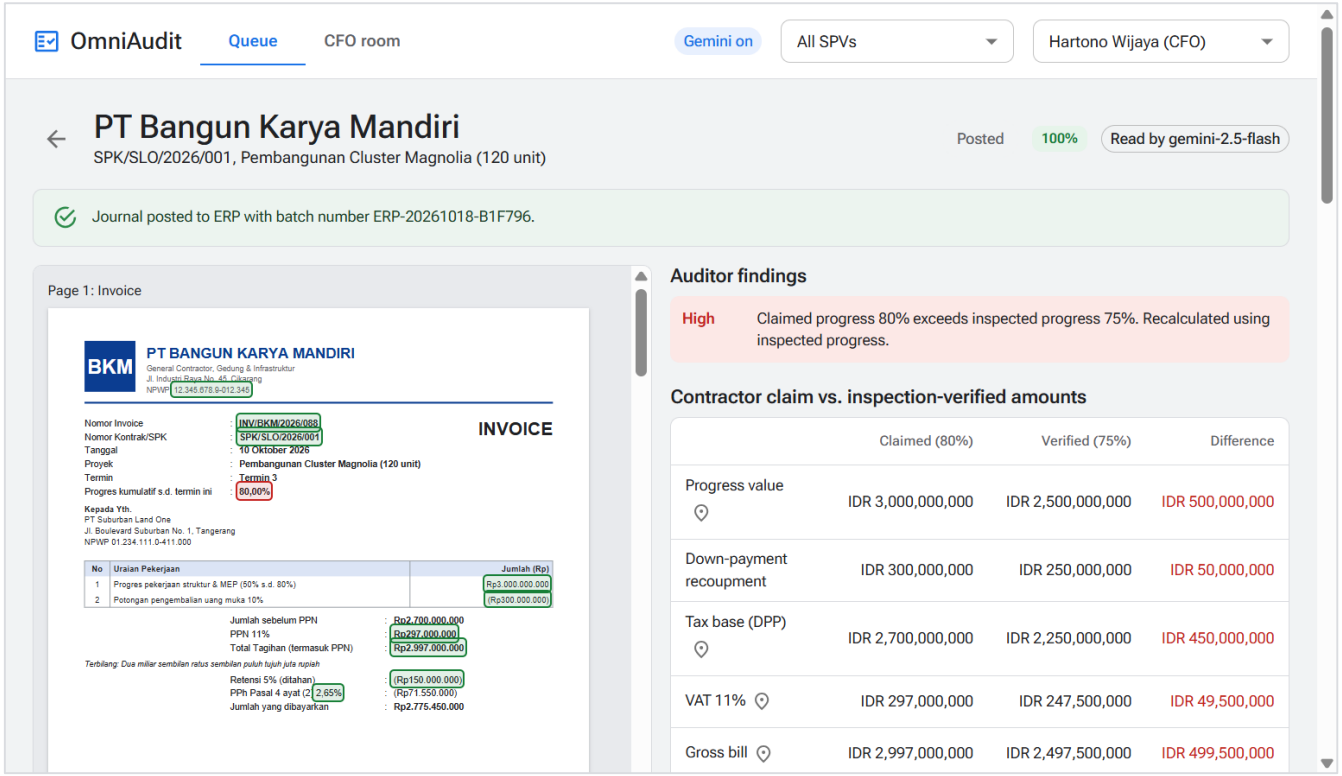


Buyer payment (receivables)



Every step writes to the hash-chained audit trail (mirrored to Firestore).

Wireframes/Mock diagrams of the proposed solution (optional)

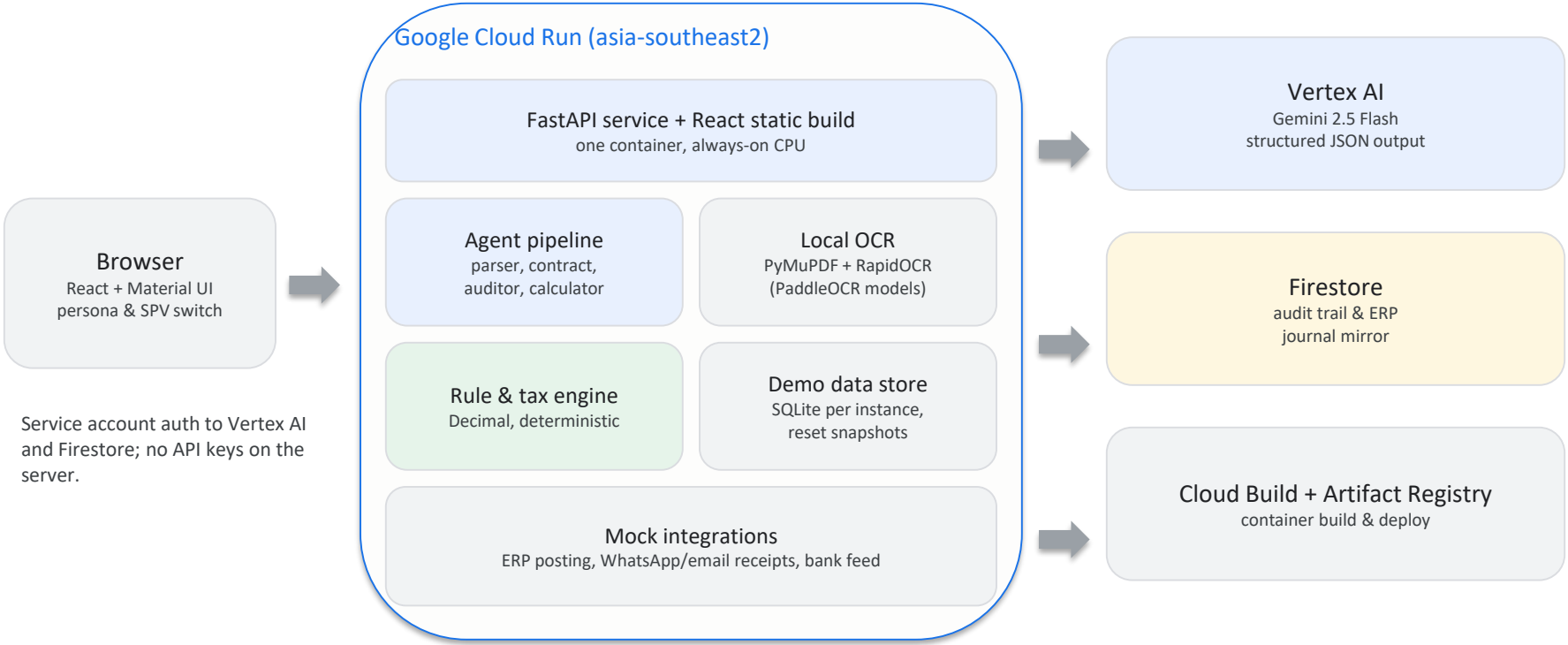


Split-view audit

Left: original PDF with highlight boxes from OCR coordinates; red marks values behind a finding.

Right: claimed vs verified amounts, findings, and the live agent trail.

Architecture diagram of the proposed solution



Technologies to be used in the solution

Gemini 2.5 Flash on Vertex AI	**Page classification and field extraction with a JSON schema; values grounded to OCR words**
Google Cloud Run	Single container for API + web app, Jakarta region, always-on CPU for background processing
Cloud Firestore	Durable mirror of the audit trail and posted ERP journals
Cloud Build / Artifact Registry	Source-to-container builds for each deploy
PyMuPDF + RapidOCR	Local OCR with word coordinates for evidence boxes; page images never leave the server
FastAPI, SQLModel, Python Decimal	Agent pipeline, rule engine and tax calculator, 80 automated tests
React, Vite, Material UI, pdf.js	Dashboard, split-view audit screen, reconciliation and CFO room

Estimated implementation cost (optional)

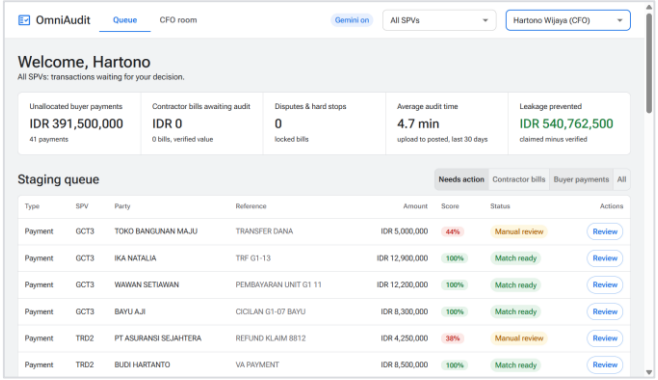
Prototype run cost (estimates, Google Cloud list prices)

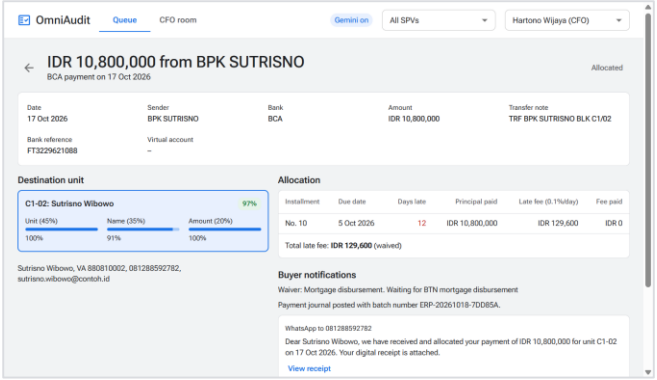
- Cloud Run, 1 instance, 2 vCPU / 2 GiB, always-on CPU: about IDR 1.7 million per month
- Gemini 2.5 Flash on Vertex AI: one request per bill (a few thousand tokens); cost scales with bill volume and is small per document
- Firestore: a few writes per action, well inside low-volume pricing
- Local OCR runs inside the container: no per-page OCR fees

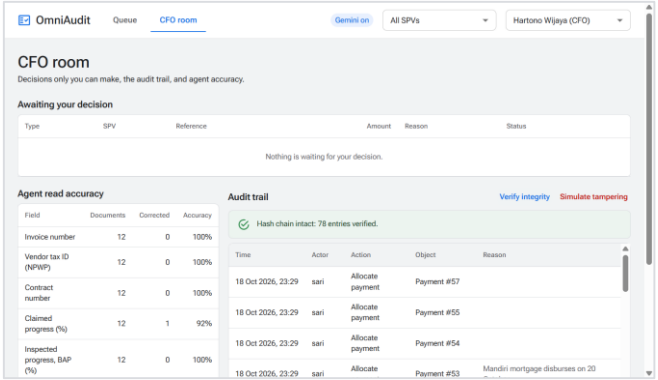
Production

Autoscaling Cloud Run, Vertex AI regional endpoint for data residency, and managed database replacing the demo SQLite store; cost driven mainly by document volume, not by user count.

Snapshots of the prototype







Dashboard

Unallocated payments, pending bills, disputes, leakage prevented.

Reconciliation

97% match to unit C1-02, late fee waived for a pending mortgage.

CFO room

Approvals, hard stops, audit trail integrity, AI accuracy.

Prototype Performance report/Benchmarking

15 / 15
full-run scenario steps pass on the live Cloud Run site with Gemini 2.5 Flash (12 bills + 3 bank statements)

98 / 98
fields matched in the Gemini extraction check on the demo documents

15–25 s
per digital bill end to end on Cloud Run; about 50 s for a scanned bill with local OCR

80
automated tests pass, including tax and journal examples that match to the rupiah

IDR 540.8M
overpayment prevented in the simulated work week (claimed minus verified)

Additional Details/Future Development (if any)

Next steps to production

- Real ERP connectors (SAP, Oracle)
- MT940 bank statements and bank webhooks
- WhatsApp Business API for receipts
- Vertex AI regional endpoint for data residency
- Retention release at final handover (BAST II)
- Pilot in one SPV with real data

Impact targets

- Audit a contractor bill in under 15 minutes instead of 3–5 working days
- No overbilling slips through; retention, penalties and down payment always deducted
- 95% of buyer payments allocated the same day

Provide links to your:

- GitHub Public Repository: https://github.com/rama2990/omniaudit-ai
- Demo Video Link (3 Minutes): [demo video link]
- Final Product Link: https://omniaudit-648878449957.asia-southeast2.run.app

One bill from the demo

Magnolia Cluster contract, progress payment 3: the invoice claims 80% progress, the site inspection says 75%, and the project is 10 days late.

IDR 2,575,450,000
requested by the contractor

IDR 2,112,875,000
after verification (progress, down payment, retention, late penalty, withholding tax)

IDR 462,575,000
overpayment stopped before payment

Because the claim is above IDR 500 million, the bill is locked and routed to the CFO. The auditor sees the 80% and 75% highlighted in the original PDF, accepts the correction, and the CFO approves; a balanced journal is posted to ERP and the decision is logged in the hash-chained audit trail.

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Thank you

